

Towards Look-Ahead Strategies for Work Item Selection

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Abstract—We introduce look-ahead strategies to sort the work-items in work lists of process agents in order to improve key performance indicators for the execution of business processes. In contrast to well known strategies relying solely on local information like first-in first-out (FIFO) or earliest-deadline-first (EDF), our look-ahead approach utilizes structural, organizational, empirical, and current workload information of the process to provide a heuristics for computing recommendations for actors, which work-item should be selected next for execution. We describe the look ahead strategies and report and discuss the results of simulation experiments for checking the feasibility of the approach.

Index Terms—Workflow, Workflow Time Management, Work list ordering,

I. INTRODUCTION

Workflows [1] are used in many enterprises to control and drive business processes. Business process management [2]–[4] comprises of many aspects; a particular aspect is to represent temporal requirements for the execution of business processes and the generation and communication of time-related information to process participants, such that they can make good decisions to improve on key performance indicators [5], [6].

In this perspective, the correct and comprehensive modeling of temporal aspects plays a crucial role. Temporal aspects of business process modeling include the formulation of temporal constraints (such as durations, deadlines, maximum time span between events) in process models, the checking of properties of temporally constrained process models, the computation of schedules for executing or enacting process instances, the checking whether process instances conform with temporal constraints, computing activity deadlines when dispatching activities to actors, pro-active avoidance of violations of temporal constraints, etc. Extensive research has been conducted in temporal issues of process modeling and execution in the last decades [7]–[21].

One important quality aspect of processes is to deliver results as quick as possible or within defined deadlines (due dates) as high response times and deadline violations will result in customer dissatisfaction or expensive penalties. Time-related process performance can be measured with various performance key figures, e.g., the turn-around time, the throughput, or the number of deadline violations. To improve these key

performance figures is one of the major concerns in workflow management and business process management [7]. Work list ordering or scheduling is one way to achieve this, by trying to find a *better* execution order of work items within the work list of participants [22]. Worklist ordering based on heuristics and (simple) strategies has the advantage that it does not require complete knowledge of the process environment as is for example necessary for shop-floor style scheduling techniques [10], [23]

In workflow systems tasks (or activities) are the steps of a process which must be assigned to agents for execution. Agents are resources, like users, machines, applications. This is done by creating new work items which are placed in work lists of an agent. In the scenarios we examined one agent can only execute one task at a time, thus the work list is essentially a queue or sequence of work items to be executed. When a task is executed and finished by an agent the referring work item is removed from her work list and a new work item that refers to the predecessor task is created in the work list of the responsible agent.

A work list ordering strategy (or work list selection strategy) basically tells the user (or the system) which item should or must be selected next for execution. Often quite simple strategies are applied, like first-in-first-out (FIFO), earliest-deadline-first (EDF) if deadlines must be met, or priority-based strategies if certain processes or tasks are more important than others. The undisputable advantage of these strategies is, that very little information is required for a selection decision, which makes them very easy to implement. E.g., for FIFO only the arrival time and for EDF only the attached deadline is required. Which strategy one uses depends on the use case and the performance goals.

In this work we aim to contribute to improve worklist selection strategies by using additional information. We assume that with additional knowledge about the current situation (e.g., workload, expected task durations, successor tasks) it is possible to look-ahead (LA) in order to make better selection decisions. The contribution of this paper is the introduction of such a look ahead strategy together with a first experimental check of the feasibility of such a look-ahead strategy for work list ordering that can be applied to improve key performance figures. We evaluated and compared our approaches against

FIFO and EDF by simulation to answer the following questions:

- Effectiveness: can LA improve time-related performance key figures?
- Robustness: is the effectiveness of LA dependent on the workload?
- Tradeoff: as the implementation of LA is not trivial, does the additional effort pay off?

The remainder of the paper is organized as follows: In section II we motivate the introduction of look-ahead strategies with an example. In section III we present some look-ahead strategies in detail. In section IV we describe some simulation experiments which are performed to compare look-ahead strategies with the traditional strategies FIFO and SDL. A discussion of these experiments is given in section V, and in section VI we draw some conclusions.

II. A MOTIVATING EXAMPLE

Figure 1 shows a BPMN [24] process model describing the steps necessary to complete the process: after task A1, either the path via B1 or D1 is taken to finish the process. At 11:00, as the work lists of agent A and agent D are empty, agent B is responsible for work items B1.1 (referring to the process task with the same name prefix B1) and B2.1 (referring to B2), and agent C for C1.1 and C1.2 (referring to C1 and another C1, each belonging to a different process instance). The expected execution duration of each task is denoted in angle brackets. Attached to each work item is the arrival time when it entered the work list and a process deadline, which tells the agent when the process the work item belongs to must be finished! When a work list contains more than one work item the agent or the workflow system has to decide which one to execute next, which is the case for agents B and C.

A. Applying FIFO

Figure 2 shows how the scenario evolves when FIFO is applied. **At 11:00:** agent C selects C1.1 for execution which is finished and removed at 13:00 (which also finishes the process in time); B selects B1.1 which is finished and removed at 13:00, and, as task C1 follows task B1 in the process model, a new work item C1.3 (arrival time 13:00, deadline 13:00) is created in the work list of C. **At 13:00:** agent C selects C1.2 for execution which finishes the process in time after two hours. Agent B selects B2.1 which is finished at 15:00 and creates item A2.1 in work list A. **At 15:00:** agent A selects and finishes A2.1 at 16:00, which violates the process deadline of 14:00. Agent C finishes C1.3 at 17:00 and also violates the process deadline of 15:00.

B. Applying EDF

As deadline violations usually result in some form of penalty, EDF seems to be the right choice for this scenario as it tries to minimize the number of deadline violations. **At 11:00:** although based on earliest deadlines the decisions are the same as in the FIFO-scenario. A difference is made **at 13:00**, as agent C selects C1.3 (instead of C1.2) which immediately

violates the process deadline, and results in another deadline violation for C1.2 **at 15:00**. As in the FIFO-scenario agent B selects B2.1 at 13:00 which results in a deadline violation for A2.1 at 14:00. Surprisingly, EDF produces one more deadline violation than FIFO.

C. Applying look-ahead

In both scenarios, the decision of B seems to be suboptimal as it is bound to result in deadline violations. The reason for that is quite obvious. **At 11:00** B has to choose his next work item: B1.1, where the successor task is C1, or B2.1 where the successor task is D2. The work list of C already contains 2 work items and agent C will most likely be busy for the next 4 hours, whereas the work list of D is currently empty, meaning that agent D is idle. By "looking ahead" only one step in the process, B can gather information that allows him to choose B2.1 over B1.1. Thus, after finishing B2.1, **13:00** agent A can start to work on A2.1, and in contrast to the other scenarios at this point no agent is idle, which ultimately results in earlier process completion times and only one deadline violation **until 15:00**. As one can easily see, applying look-ahead in this scenario produces less deadline violations, higher resource utilization, and decreased average turnaround.

III. LOOK-AHEAD STRATEGIES

Our lookahead-approach requires and utilizes additional information: the process model (successor tasks), organizational information (agent assignments), empirical data or forecasts (expected activity durations and branching probabilities), and the current workload (resource utilization, current load on work lists).

In our example each task was assigned to exactly one agent, in real world scenarios a work item can be selected and executed by a currently idle agent (resource) out of a group of agents (resource pool) with the same qualification (function or role).

Definition 1 (Agents): An agent $a \in A_q \subset A$ is a member of exactly one group of agents with the same qualification q , such that $a \in A_q \implies a \notin A_r, r \neq q$. The capacity $|A_q|$ depicts the number of agents in the group q .

Definition 2 (Tasks): A task $t = (A_q, d)$ is assigned to a group of agents A_q responsible for the execution of t , and has an expected duration $0 < d \in \mathbb{R}$.

Definition 3 (Simplified Process Model): A process model $P = (T, E, A, \delta)$ is defined by a set of tasks $t \in T$, a set of directed edges $e \in E$ connecting the tasks, a set of agent groups $A_q \in A$ and maximum process duration δ to calculate a deadline for the process. Each edge $e = (t_1, t_2, p)$ describes a predecessor-successor relation between two tasks $t_1 \in T$ and $t_2 \in T$, where p is the probability that t_2 will be executed when t_1 is finished.

To simplify and focus our explanations we omitted all control nodes and restrict structures to sequences and conditional branches, which excludes parallel execution. The start task is the one task without predecessors (A_1 in our example), and an end task is one without successors (C_1 and D_2). The xor-split

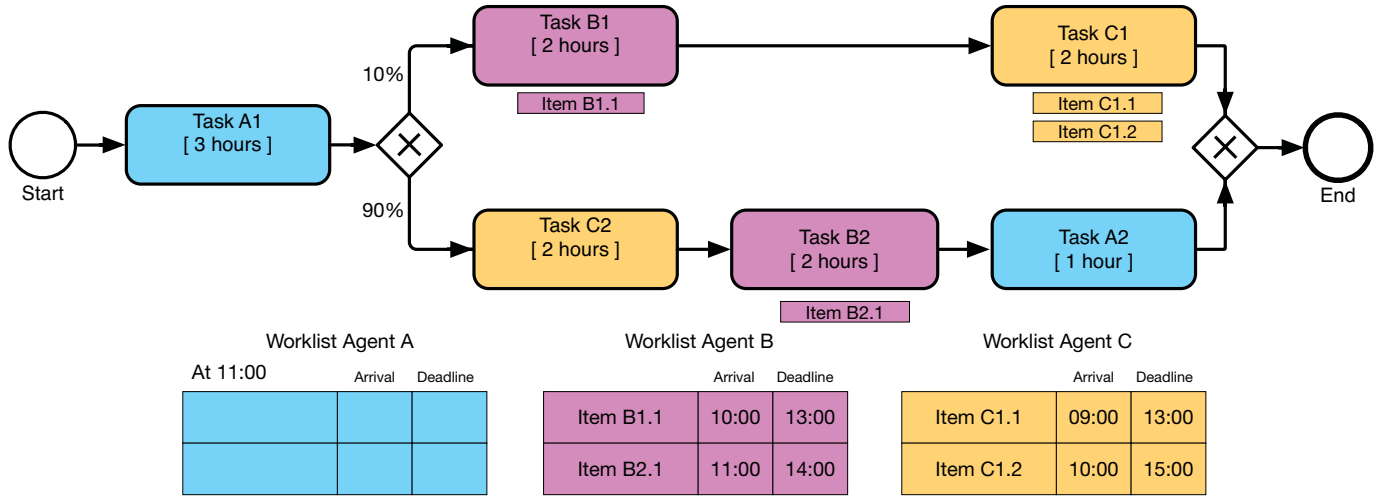


Fig. 1. Example process scenario with work lists

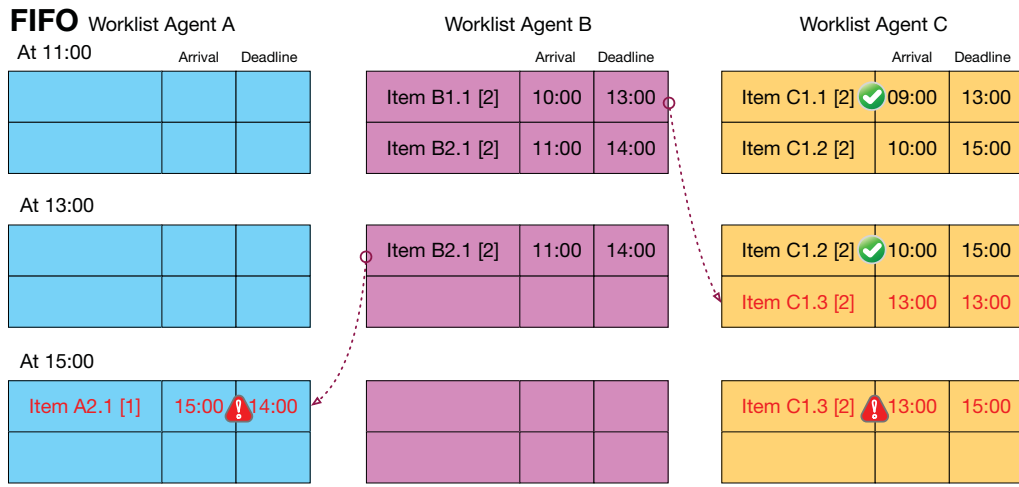


Fig. 2. Scenario with FIFO resulting in 2 deadline violations

is represented by the edges $(A1, B1, 0.1)$ and $(A1, D1, 0.9)$. For a complete model with additional explanations please refer to [25].

For reasons of convenience the attributes of tuples can also be written in a dot-notation, e.g., for a task $t = (A_q, d)$ the agent group can also be denoted as $t.A_q$ and the duration as $t.d$.

Definition 4 (Worklist): Each group of agents A_q owns a work list $W_q = \langle w_1 \dots w_n \rangle$, which is a sequence of n work items $w_i, 0 \leq i \leq n$ to be executed by an agent $a \in A_q$. $|W_q| = n$ depicts the number of work items in the list of agent a . A work item $w_i = (t, arr, due)$ refers to the task $t \in T$ which arrived in the work list at time arr and has a process deadline of time due attached to it. The function $wl(t) \in W$ returns the work list $W_a, t.a = a$. of the agent a responsible for task t .

A. Weighted Time and Resource Based Look-ahead (WTR)

WTR orders the work list based on the weighted utilization of resources that are responsible for the execution of the succeeding tasks. The utilization $util_t$ for one particular task $t = (A_q, d)$ is equal to the work load $load_q$ of the assigned agent group A_q , which is determined by the expected sum of durations of all work items in their work list $w_i \in W_q$, where $w_i = (t_i, arr_i, due_i)$ refers to a task t_i , and their capacity $|A_q|$:

$$util(t) = load(q) = \frac{\sum t_i.d}{|A_q|} \quad (1)$$

For instance $util(B1)$ at 11:00 to be executed by one agent B is calculated as follows:

$$util(B1) = \frac{B_{1.1}.d + B_{2.1}.d}{|B_q|} = \frac{2 + 2}{1} = 4$$

$$util(C1) = \frac{C_{1.1}.d + C_{1.2}.d}{|C_q|} = \frac{2 + 2}{1} = 4 \quad (2)$$

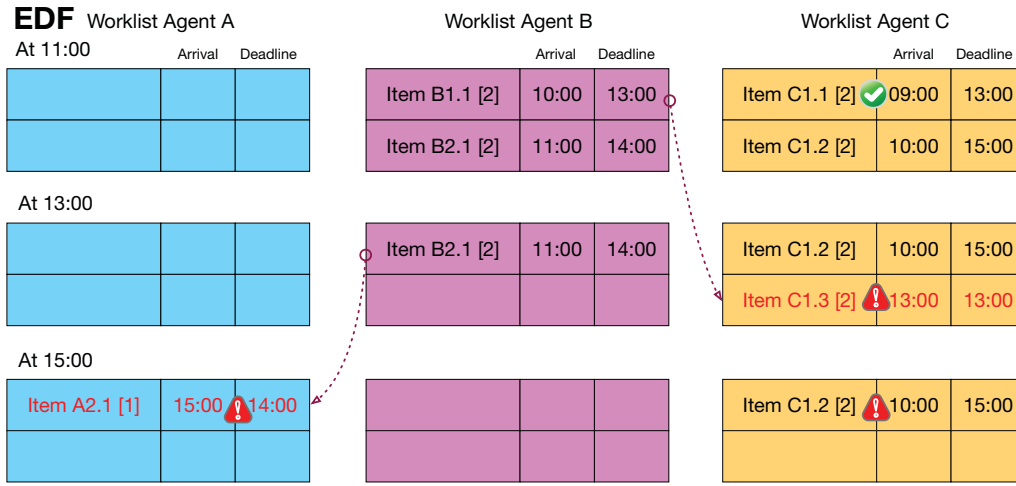


Fig. 3. Scenario with EDF resulting in 3 deadline violations

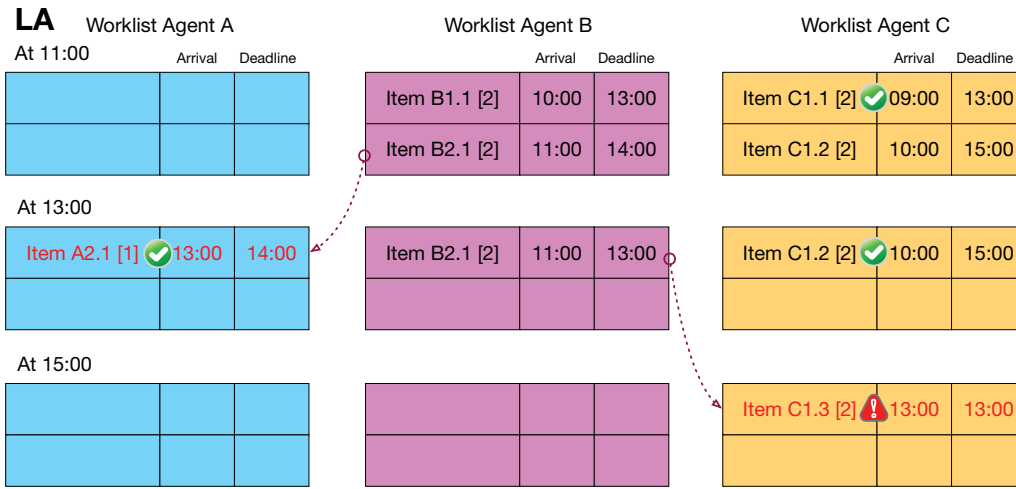


Fig. 4. Scenario with LA resulting in 1 deadline violation

The 0-look-ahead value for a task t (that refers to a particular work item) is equal to the utility value of the task itself.

$$la(t, 0) = util(t) \quad (3)$$

The k -look-ahead value, for $k > 0$, for a task t is the combined and weighted utility value of k transitive successors, determined by the task successors s_i defined by edges $(t, s_i, p_i) \in E$.

$$la(t, k) = la(t, 0) + \sum p_i * lat(s_i, k - 1) \quad (4)$$

Now we can use this to order work lists and select the next item to execute, which is the one with minimum look-ahead value. For instance the 1-look-ahead of the work list of B at 11:00 containing work item $B1.1$ that refers to task $B1$ and the $B2.1$ that refers to $B2$ are:

$$\begin{aligned} la(B1, 1) &= util(B1) + (1.0 * util(C1)) = 8 \\ la(B2, 1) &= util(B2) + (1.0 * util(A2)) = 4 \end{aligned} \quad (5)$$

Accordingly $B2.1$ would be selected. The following example shows how to calculate the 2-look-ahead with xor-splits and weighted utilization: assume that a new process is started at 13:00 in our LA-scenario, which creates an additional work item $A1.1 = (A1, 13:00, 19:00)$ in work list A. At this point in time $load(A) = 3 + 1 = 4$, $load(B) = 2$ and $load(C) = 2$, thus we calculate the look-ahead for $A1.1$. as follows:

$$\begin{aligned} la(A1, 2) &= util(A1) \\ &+ 0.1 * (util(B1) + (1.0 * util(C1))) \\ &+ 0.9 * (util(C2) + (1.0 * util(B2))) \quad (6) \\ &= 4 + 0.1 * 4 + 0.9 * 4 = 4 \end{aligned}$$

As one can see we weight the look-ahead values of the successors of $A1$ according to the branching probabilities defined on the edges $(A1, B1, 0.1)$ and $(A1, D1, 0.9)$.

B. Deadline-oriented Weighted Time and Resource Based Lookahead (DWTR)

Like FIFO, and unlike EDF, our WTR-strategy does not take deadlines into consideration, thus 'late' work items can not be prioritized. Thus we developed DWTR which adapts WTR and additionally takes the probability of an upcoming deadline violation into account, which we can determine by applying the most-probable-deadline-violation algorithm (MPDV) we presented in [12]. MPDV utilizes branching-probabilities and task durations to calculate a deadline-violation-probability for each work item.

DWTR integrates MPDV into a two-step work list ordering algorithm:

- 1) Sort work items in descending sort-order by deadline violation probability determined by MPDV. The work item with the highest probability will be executed first, or in other words: already late processes will be prioritized.
- 2) If two items have the same deadline violation probability apply WTR to select the one with the lower look-ahead value.

E.g., for a work item $w = (B1, 10:00, 12:00)$ the probability of a deadline violation is 1.0 (100%) as the process will be finished at 14:00 (2 hours for B1 + 2 hours for C1) and the deadline is 12:00. For a work item $w = (A1, 10:00, 14:00)$ the probability of a deadline violation is 0.9 (90%), as there is a 90%-chance that it will be finished too late at 15:00, if the path via C2 is chosen. On the path via B1 the process will be finished at 14:00, which implies that there is a 10%-chance that the process will finish within the deadline.

IV. EVALUATION

The goal of our evaluation efforts was to test the effectiveness and robustness of our strategies in comparison to the traditional selection strategies FIFO and EDF. We developed a business process simulation environment based on jBPM [26] and simulated different scenarios [25] based on process models with deadlines, agent assignments, empirical branching probabilities, empirical task durations, and variations of the following start parameters: process arrival times (frequency of process starts), look-ahead and aging-factors.

We performed simulation-runs on various complex scenarios and logged run-time information about the execution of work items to calculate the following performance indicators: average resource utilization (how busy or idle were agents), turnaround (average execution time of all processes), number of started processes, throughput (number of finished processes), number of deadline violations of finished processes, number of deadline violations of still active (possibly starving) processes, and the tardiness (degree of deadline exceeding).

Here we only present a summary of the major findings, more details on simulation results and interpretations can be found in [25]. In comparison to FIFO and EDF, we achieved performance improvements for some scenarios, but also performance deterioration for others.

A. FIFO versus WTR

For FIFO versus WTR we focused on the performance indicators 'turnaround time (cycle time)' and 'throughput', which revealed the following strong tendencies.

- **Robustness:** WTR seems to be robust; in most scenarios, from very low to very high resource utilization, we registered improvements on the performance indicators.
- **Effectiveness:** WTR performed better than FIFO in most scenarios; the degree of improvement over these scenarios varied heavily, between 5 % to 50%. Very unexpected was that a look-ahead greater than one did often not result in better performance indicators.
- **Tradeoff:** considering the experienced improvements, the additional effort of implementing WTR seems to pay off; currently we propose to implement a look-ahead of 1.

B. EDF versus DWTR

For the comparison of EDF and DWTR we focused on the performance indicators 'number of deadline violation'. Again we could find some strong tendencies:

- **Robustness:** in (underloaded) scenarios with very low resource utilization there is not much effect, as agents often have only one work item to choose from. In (balanced) scenarios with average resource utilization and deadlines that are not too tight, the number of deadline violations can be significantly decreased with DWTR compared to EDF. In (overloaded) scenarios with high resource utilization we observed that rising resource utilization implies a higher number of deadline violations.
- **Effectiveness:** the effectiveness of DWTR varied heavily depending on resource utilization. Again, a look-ahead greater than one, surprisingly, did often not result in better performance indicators.
- **Tradeoff:** considering the robustness-result an implementation and application of DWTR is heavily dependent on the scenario and expected load. We suggest to simulate the scenario before implementing it in a workflow system.

An analysis of the robustness problems in DWTR showed that an overloaded system that prefers already late work items will produce even more late work items, although we could not really explain why EDF performed better. We tried to counter this by introducing an aging-factor, usually used to prevent starving. In our case each work item has a selection counter that is increased when another work item is chosen for execution. When this counter reaches the value defined by the aging-factor the work item can no longer be ignored. However, the aging-factor did not bring the desired improvements.

V. DISCUSSION

The evaluation of look-ahead strategies vs. other strategies by simulation produced promising results, especially for WTR which can improve turnaround times and throughput for most tested scenarios. Nevertheless, these first results also showed that we currently do not fully understand why certain effects occur, for instance why DWTR performs worse than EDF in

scenarios with a high system workload or why a look-ahead deeper than 1 did not improve the results in many scenarios. We suspect that the decision to choose one particular work item to be executed next may in certain types of scenarios have unforeseeable feedback effects on other work items, like speeding up one item may slow down several others, or freeing one participant may result in overloading another participant.

However, this was a first expedition into the very promising research area of look-ahead strategies for work list sorting, opening several avenues for further exploration. In particular, we will examine the above mentioned feedback effects in order to acquire a deeper understanding of cause and effects. Additionally, this will help us to categorize different types of workflow scenarios, resource assignments, and workload variants that will allow us to develop new look-ahead strategies, as well as the possibility of dynamically switching between different strategies at run time.

VI. CONCLUSION

The selection of work-items from work-lists belongs to the most important decisions for the performance of business process execution. In this paper we introduced the idea of using look-ahead strategies for work list sorting, respectively for selecting work-items from worklist. We presented two look-ahead strategies: the Weighted-Time-and-Resource-based look ahead (WTR) and the Deadline-oriented WTR (DWTR). In contrast to well known strategies like first-in-first-out (FIFO) and earliest-deadline-first (EDF), our new strategies utilize additional information like successor tasks and agent assignments, expected activity durations and branching probabilities, and the current work load on agents.

The experimental evaluation of the different strategies showed that in several scenarios look-ahead strategies for decisions are indeed a promising alternative to more traditional strategies. However, additional research is necessary to better predict under which circumstances which heuristics performs best.

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